Safety Data Sheet

Chairside Monomer for Soft Reliner **SDS DATE:** 12/07/11

SECTION 1: PRODUCT AND COMPANY IDENTIFICATION

Product Name: Chairside Monomer for Soft Reliner

Other Names: R-CHEM-149, n-Butyl Methacrylate Solution

Product Use: Organic process chemical.

Distributor:
Septodont Confi-Dental Division
416 S. Taylor Ave.
Louisville, CO 80027
USA
Telephone: (800) 383-5158

Emergency Information Chemtrec:
(703) 527-3887

SECTION 2: HAZARDS IDENTIFICATION

Primary Routes of Entry:
Skin, Inhalation, Eyes.

Potential Health Acute Effects:

Eyes:
Irritation to the eyes including but not limited to blurred vision, watering and discomfort.

Skin:
Repeated contact with skin may cause irritation. Repeated or prolonged use can lead to more severe symptoms such as severe skin reactions, and organ damage, symptoms may be delayed.

Inhalation:
Irritation may occur. Repeated and/or prolonged exposure can cause dizziness, headache and anesthetic effects.

Ingestion:
Possible irritation and/or pain with ingestion include but are not limited to the mouth, esophagus, respiratory and digestive tracts.

Potential Chronic Health Effects:
Prolonged or overexposure at near lethal concentrations can cause organ damage.

Note to Physicians:
This product contains N,N-Dimethyl-p-Toluidine at low concentrations (refer to section 3). While complications from this component are not expected, absorption leads to formation of methemoglobin, which in sufficient concentration causes cyanosis. Symptoms may include headaches, weakness and dizziness, and can be recognized by the blue color of the lips, fingernails, nose and earlobes. Reversion of methemoglobin to hemoglobin occurs spontaneously after termination of exposure, moderate degree of cyanosis needs to be treated only by supportive measures such as bed rest and oxygen inhalation. Thorough cleansing of the entire contaminated area of the body is of utmost importance. If cyanosis is sever, intravenous injection of methylene blue, 1-2 mg/kg body weight over a 5 minute period as a 1% solution may be of value. If elevated methemoglobin persists after an hour, the treatment may be repeated, but the total dose should not exceed 7 mg/kg body weight. Cyanocobalmin (Vitamin B-12), 1 mg intramuscularly is reported to speed recovery. Intravenous fluids and blood transfusions may be indicated in very severe exposures.

Carcinogenicity:
Dibutyl Phthalate is a possible carcinogen according to the EPA and OSHA. None of the remaining components were listed by IARC, NTP, OSHA or ACGIH as carcinogens.

SECTION 3: COMPOSITION/ INFORMATION OF INGREDIENTS

COMPONENT:	CAS NUMBER:	WT/WT %:
Dibutyl Phthalate	**84-74-2**	**40.0 – 70.0**
N-Butyl Methacrylate	**97-88-1**	**40.0 – 70.0**
N,N-Dimethyl-p-Toluidine	**99-97-8**	**0.0 – 5.0**
Proprietary	**Proprietary**	**Proprietary**

SECTION 4: FIRST AID MEASURES

Eyes:

Flush with copious amounts of water for at least 15 minutes.

Skin:

If irritation occurs and product is on the skin, rinse thoroughly with water, followed by a thorough washing of the affected area with soap and water. If irritation, redness or swelling persists, contact a physician immediately.

Ingestion:

If ingested, do not induce vomiting. If product has been swallowed, drink plenty of water or milk. If the patient is vomiting, continue to offer water or milk. Never give anything by mouth to an unconscious person. Provide an estimate of the time at which the material was ingested and the amount of the substance that was swallowed. Seek medical attention immediately.

Inhalation:

Immediately seek fresh air. Open windows and/or doors to increase ventilation. Seek immediate medical attention.

SECTION 5: FIRE-FIGHTING MEASURES

Flammable/Explosive Limits:

Air Vol %: Lower: 2.0, Upper: 8.0

Flash Point:

49°C, 120°F

Method Used:

No information.

Autoignition Temperature:

294°C, 561°F

Sensitive to Mechanical Impact:

No.

Sensitive to Static Discharge:

Yes.

Extinguishing Media:

Chemical Foam, CO_2, Dry Chemical Powder, Water Spray. NOTE: Water may be ineffective in extinguishing the fire.

Special Fire Fighting Measures:

This product is a flammable liquid. When involved in a fire, this product ignites readily and decomposes to produce carbon oxides. Vapors of this product are heavier than air and may travel to a source of ignition and flash back to a leaking or open container. Do not enter fire area without proper protection. Fight fire from a safe location. Heat/ impurities may cause pressure to build and/or closed containers, spreading fire, increasing risk of burns/injuries. Structural firefighters must wear SCBAs and full protective equipment.

Unusual Fire and Explosion Hazards:

High temperatures, inhibitor depletion, accidental impurities, or exposure to radiation or oxidizers may cause spontaneous polymerizing reactions generating heat/pressure. Closed containers may rupture or explode during a runaway polymerization. Use a water spray or fog to reduce or direct vapors. Water may not be effective in actually extinguishing a fire involving this product.

SECTION 6: ACCIDENTAL RELEASE MATERIALS

Process for cleaning and take-up:

Evacuate the spill area. Maximize ventilation by opening doors and windows. Eliminate all sources of ignition. Use self-contained breathing apparatus and protective clothing, (e.g. goggles and gloves). Dike and absorb or contain the spill with an inert material, such as sand. Use non-sparking tools ONLY and dispose of the material in the proper closed container(s). Contaminated monomer may be unstable, add inhibitor to prevent polymerization.

Wash all affected areas of the skin with warm water and soap. Remove clothing that has been contaminated and wash thoroughly.

Environmental Precautions:

Keep spills and cleaning runoffs out of sewers and open bodies of water. Spills on porous surfaces can contaminate the ground water. Dispose of in accordance with Federal, State and Local regulations, or appropriate governing standards.

SECTION 7: HANDLING AND STORAGE

Handling:

Handling advice:

To be handled by qualified dental professionals.
Follow instructions for use.
After each use, wash hands or exposed skin thoroughly. Do not smoke, eat or drink while using this product.

Fire and explosion protection:

Store in a cool dry place away from heat, sparks, flame and direct sunlight. Close container after each use. Use explosion-proof equipment. Check inhibitor levels every three months.

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

Exposure Guidelines:

Chemical Name:	CAS NUMBER:	Exposure Limits in Air:	
		OSHA-PELs	ACGIH-TLVs
Dibutyl Phthalate	84-74-2	5 mg/m^3	5 mg/m^3
N-Butyl Methacrylate	97-88-1	NE	NE
N,N-Dimethyl-p-Toluidine	99-97-8	NE	NE
Proprietary	Proprietary	NE	NE

Engineering Controls:

Ventilation:

Production: use explosion-proof local exhaust at processing equipment, including buffers, sanders, grinders and polishers. High temperature processing equipment should be well ventilated.

Professional: Follow precautions and directions on label.

Personal Protection:

Respiratory Protection:

A respirator should be worn whenever workplace conditions warrant its' use. If necessary, use only respiratory protection authorized per U.S. OSHA's requirement in 29 CFR § 1910.134 or other appropriate governing standard.

Hand Protection:

Chemical resistant gloves should be worn when prolonged or repeated skin contact is expected. If necessary, refer to U.S. OSHA 29 CFR § 1910.138, or other appropriate governing standard. Eye wash stations, safety showers or sink basins are recommended.

Eye Protection:

Safety glasses or chemical splash goggles may be worn, if necessary refer to U.S. OSHA 29 § 1910.133 or other appropriate governing standard. Eye wash stations, safety showers or sink basins are recommended.

Other:

No special body protection is required, refer to appropriate governing standard(s).

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

General description:

State: Oily Liquid
Odor: Acrid, Fruity
Color(s): Clear, Pale

Chemical Properties:

Designation:	Value:

Vapor Pressure	4.9 mm Hg at 20º C/ 68º F
Boiling Point **Freezing Point**	163º C, 325º F ND
Solubility in Water	Negligible
Viscosity	0.92 cps at 25º C/ 77º F
Specific Gravity (H_2O = 1)	0.889 g/cc at 25º C/ 77º F
Percent Volatile W/W%	<70%
Vapor Density (AIR = 1)	>1
Evaporation Rate (BuAc = 1)	<1

SECTION 10: STABILITY AND REACTIVITY

Stability:

This material is stable until depletion of inhibitor; product will become unstable or reactive.

Reactivity:

Hazardous Decomposition Products: Oxides of carbon when burned.

Incompatibility: Heat and ignition sources, aging and contamination. Strong oxidizers, strong reducers, free radical initiators, inert gases, oxygen scavengers.

Hazardous polymerization: Polymerization may occur.

Other: Avoid high temperatures, localized heat sources, oxidizing conditions, freezing conditions, direct sunlight, ultraviolet radiation, inert gas blanketing. Avoid spills or applications to paint and rubber due to solvent like properties that will cause damage.

Section 11: Toxicological Information

General toxicological information:

Dibutyl Phthalate: Kidneys, Central Nervous System, Eyes, Male Reproductive System
Methacrylate: Nervous system
N,N-Dimethyl-p-Toluidine: Liver, Central Nervous System, Blood and Skin
For proprietary: None listed.

Toxicity Data:

Dibutyl Phthalate:		
Acute Oral (Rat)	LD_{50}:	7000 mg/kg
Intraperitoneal (Rat)	LD_{50}:	6889 µL/kg
Methacrylate:		
Inhalation (Rat)	LC_{50}:	>2900 ppm – 8H
Oral (Rat)	LD_{50}:	>5000 mg/kg
Skin (Rabbit)	LD_{50}:	>2900 mg/kg
N,N-Dimethyl-p-Toluidine:		
Inhalation (Rat)	LC_{50}:	254 ppm/ 4H
Acute Dermal (Rat)	LD_{50}:	>2000 mg/kg
Ingestion (Rat)	LD_{50}:	1650 mg/kg

Trade Secret:		
Oral (Rat)	TD_{LO}:	45 gm/kg
Oral (Rat)	LD_{LO}:	54 gm/kg
Intraperitoneal (mouse)	LD_{50}:	300 mg/kg
Oral (Rat)	LD_{50}:	7400 mg/kg

SECTION 12: ECOLOGICAL INFORMATION

General ecological information:

Aquatic toxicity is possible, including physical damage or death to aquatic life; do not discharge into drains or the environment.

Dibutyl Phthalate:		
Flathead Minnow	LC_{50}:	0.92 mg/L/96H

SECTION 13: DISPOSAL CONSIDERATIONS

Waste disposal of product:

This product contains a Phthalate (UO69) and a Methacrylate Monomer; incinerate liquid and diking material, after addition of excess inhibitor, in accordance with Federal, State, and Local regulations or appropriate governing regulations.

Do not reuse empty drums or containers due to potential hazards. Dispose of all empty containers according to Federal, State, and local or appropriate governing regulations.

SECTION 14: TRANSPORT INFORMATION

Proper shipping name: RQ, Flammable Liquid, NOS
Hazard class or division: 3
Identification number (UN): 1993
Packing group: III

The transport information provided represents the regulatory transport classification of the product without consideration to packaging, quantity, or modal restrictions and exceptions. It is the user's responsibility to determine the appropriate packaging and modal requirements and/or limitations for the product quantity being shipped.

SECTION 15: REGULATORY INFORMATION

Symbols of Danger:

None

R-Phrase:

R10- Flammable
R36/37/38 – Irritating to eyes and respiratory system and skin.
R43 – May cause sensitization by skin contact.
R50- Very toxic to aquatic systems.
R61- May cause harm to unborn baby.
R62- Possible risk of impaired fertility.

S-Phrase:

S3 – Keep in cool place.
S7 – Keep container tightly closed
S9 – Keep container in well ventilated place
S16 – Keep away from sources of ignition – No Smoking.
S20 – When using do not eat or drink
S29 – Do not empty into drains
S37/39 – Wear suitable gloves and eye/face protection.
S45- In case of accident or if you feel unwell, seek medical advice immediately (show label where possible).
S53- Avoid exposure – obtain special instructions before use.
S61- Avoid release into the environment. Refer to special instructions/ Safety Data Sheets.

SECTION 16: OTHER INFORMATION

Hazardous Material Identification System (HMIS) Rating:

Health:	1
Flammability:	2
Reactivity:	2
Personal Protective Equipment:	Gloves and Safety Glasses or Chemical Splash Goggles

National Fire Protection Association (NFPA) Hazard Identification Rating:

Health:	1
Flammability:	2
Reactivity:	2

Warning Codes:

Xi – Irritant
T – Toxic
N – Dangerous to the Environment

All information, recommendations, and suggestions appearing herein concerning our product are based upon tests and data believed to be reliable. However, it is the user's responsibility to determine the safety, toxicity, and suitability for his own use of the product described herein. Since the actual use by others is beyond our control, no guarantee, express or implied, is being made as to the effects of such use, the results obtained, or the safety and toxicity of the product nor is there any assumed liability arising out of use, by others, of the product referred to herein. The information herein is not to be construed as absolutely complete since additional information may be necessary or desirable when particular or exceptional conditions or circumstances exist or because of applicable laws or government regulations.